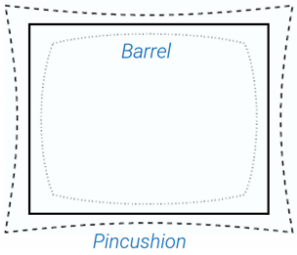
Lens Distortion and Radial Distortion in Image Formation

In the image formation process, **lens distortion** occurs due to imperfections in the lens that cause light rays to bend in ways that do not perfectly match the ideal pinhole camera model. This results in straight lines appearing curved, which is noticeable especially towards the edges of the image.

- **Radial lens distortion** is one of the most common types of distortion. It happens because light rays passing through the lens are bent by varying amounts, depending on how far they are from the center of the image. This results in two types of distortion:
    - **Barrel distortion**: Straight lines bend outward, causing the image to appear "pushed out" from the center.
    - **Pincushion distortion**: Straight lines bend inward, making the image look like it's pulled in toward the center.



Tangential distortion Second-order effects caused by misalignment or defects of the lens (i.e. capture distortions that are not considered in radial distortion).

Modeling lens distortion

Lens distortion can be modeled using a non-linear transformation that maps ideal (undistorted) image coordinates $(x_{\text{undist}}, y_{\text{undist}})$ into the observed (distorted) coordinates (x, y):

$$\begin{bmatrix} x \\ y \end{bmatrix} = \underbrace{L(r) \begin{bmatrix} x_{\text{undist}} \\ y_{\text{undist}} \end{bmatrix}}_{\text{Radial distortion}} + \underbrace{\begin{bmatrix} dx(x_{\text{undist}}, y_{\text{undist}}, r) \\ dy(x_{\text{undist}}, y_{\text{undist}}, r) \end{bmatrix}}_{\text{Tangential distortion}}$$

where:

- r is the distance from the distortion center which is usually assumed to be the piercing point $c = (0, 0)$. Therefore, $r = \sqrt{(x_{\text{undist}})^2 + (y_{\text{undist}})^2}$.
- $L(r)$ is the radial distortion function which is a linear operator defined for positive r only and is approximated using the Taylor series:

  $$L(0) = 1 \qquad L(r) = 1 + k_1 r^2 + k_2 r^4 + k_3 r^6 + \dots$$

  where k_i are additional intrinsic parameters.
- The tangential distortion is approximated as:

  $$\begin{bmatrix} dx(x_{\text{undist}}, y_{\text{undist}}, r) \\ dy(x_{\text{undist}}, y_{\text{undist}}, r) \end{bmatrix} = \begin{bmatrix} 2p_1 x_{\text{undist}} y_{\text{undist}} + p_2(r^2 + 2(x_{\text{undist}})^2) \\ 2p_1 y_{\text{undist}} x_{\text{undist}} + p_2(r^2 + 2(y_{\text{undist}})^2) \end{bmatrix}$$

  where p_1 and p_2 are additional intrinsic parameters.

| **Remark.** This approximation has been empirically shown to work.

The radial distortion coefficients $k1$, $k2$, …,kn , together with the two tangential distortion coefficients $p1$ and $p2$ form the set of the lens distortion parameters, which extends the set of intrinsic parameters required to build a realistic camera model.

(B) Initial Guess for Radial Distortion Parameters in Zhang's Algorithm

In **Zhang's algorithm** for camera calibration, radial distortion is estimated as part of the intrinsic calibration process. The goal is to refine the camera parameters (intrinsics, rotation, translation, distortion parameters) by minimizing the reprojection error.

How the Initial Guess for Radial Distortion Parameters is Obtained:

1. **Collect a set of known 3D world points and their corresponding 2D image points** (from multiple images of a planar checkerboard pattern).
2. **Compute an initial estimate for the camera's intrinsic parameters**, including the focal length and principal point (using DLT or other methods).
3. **Solve for the radial distortion parameters**:
    - To obtain an initial guess for the radial distortion parameters, Zhang's algorithm models the distortion effect as part of the overall camera matrix using the distortion function mentioned above.
4. **Formulate a linear system**:
    - The system involves finding the values of the distortion parameters k1,k2,k3k_1, k_2, k_3k1,k2,k3 that minimize the error between the projected 3D points and their observed 2D projections.
    - The system is typically nonlinear (due to the presence of the radial distortion terms r2,r4,r6r^2, r^4, r^6r2,r4,r6) but is linearized by using an iterative optimization method.
5. **Linear System Setup**:
    - After obtaining the camera matrix AAA (intrinsic matrix), you can use it to map the 3D world coordinates onto the 2D image plane.
    - The residual errors between the predicted image points and the actual observed image points are used to solve for the distortion parameters.
6. **Solution Method**:
    - A **least squares** optimization is used to solve for the distortion parameters. In practice, this is often done using **Levenberg-Marquardt** or **Gauss-Newton** methods.
    - The system involves matrices representing the relationship between the known 3D world coordinates, the camera intrinsic matrix, and the observed 2D image points.

(C) Undistortion of Images After Calibration

Once the **camera calibration process** is complete and the **distortion parameters** are estimated, we can **undistort** the images. The reason for undistorting images is to remove the lens distortion effects (radial distortion, in particular) and restore the original geometric relationships in the scene, which is important for tasks like 3D reconstruction, object recognition, or precise measurements.

1. Undistort Warping

- **What is it?** After calibrating a camera using Zhang's method and estimating the lens distortion parameters (such as radial distortion), we can "undistort" an image, removing the distortion introduced by the lens.
- The distortion correction is applied using a **backward warping** approach.
- **How is it done?** The un-distortion process uses a formula to adjust the distorted image points back to their true, undistorted positions. The formula for the distortion correction in the x and y directions is:

$$u_{\text{undist}} = u_{\text{undist}} + (k_1 r^2 + k_2 r^4)(u_{\text{undist}} - u_0)$$

$$v_{\text{undist}} = v_{\text{undist}} + (k_1 r^2 + k_2 r^4)(v_{\text{undist}} - v_0)$$

- **Explanation**:
  - u_{undist} and v_{undist} are the original coordinates of a point in the distorted image.
  - $r^2 = u_{\text{undist}}^2 + v_{\text{undist}}^2$ is the radial distance of the point from the center of the image.
  - k_1 and k_2 are the radial distortion coefficients.
  - u_0 and v_0 are the image center coordinates.

Why do we need this?

- **Goal**: After calibration, we need to adjust the image to remove the distortion, so that straight lines in the real world remain straight in the image.